# Convergence of a Fourier series

We learned how to find the Fourier series representation of a function, but we haven't talked yet about whether or not the series actually converges to the function.

Remember, just like with the Taylor series we looked at before, or with any other series representation, the series doesn't necessarily converge. In the case of the Fourier series representation, we're looking for the series to converge to the function $f(x)$ that we're claiming it represents.

Luckily for us, there are certain conditions under which we know the Fourier series representation will, in fact, converge to $f(x)$.

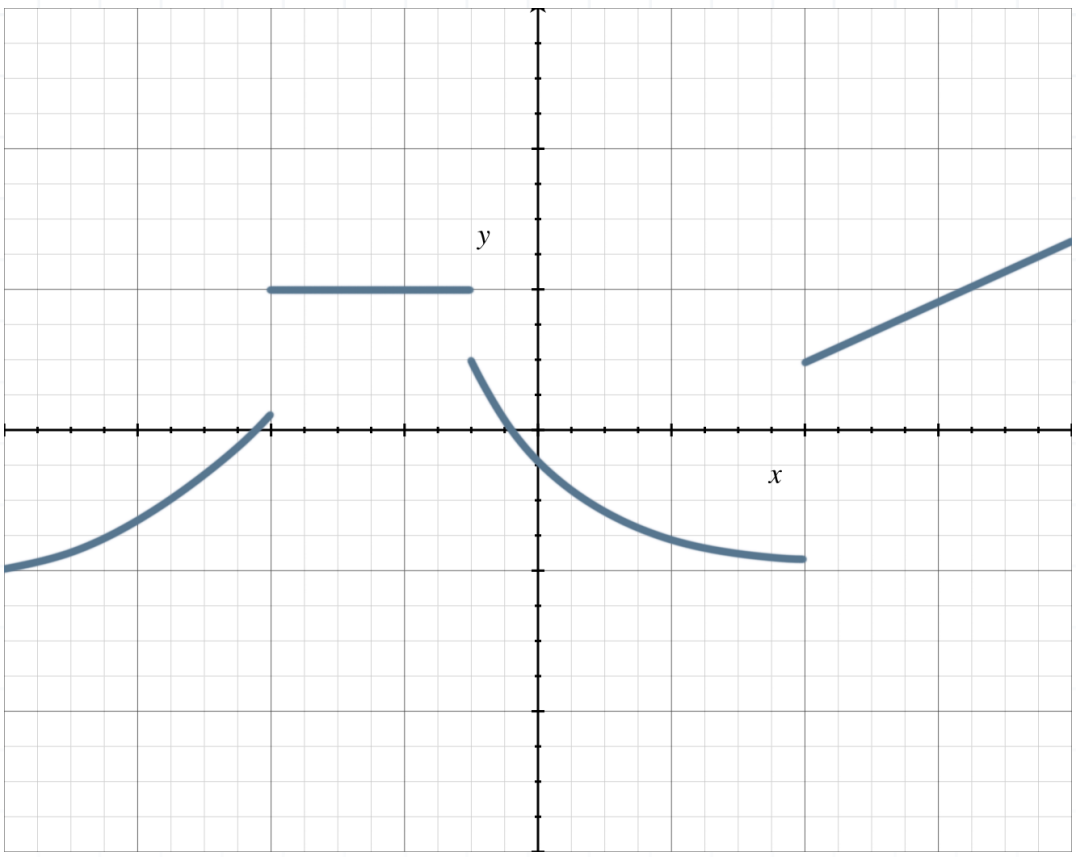
## Conditions for convergence

In order for the Fourier series representation of $f(x)$ to converge on some symmetric interval $-L \leq x \leq L$, the function $f(x)$ itself must be **piecewise smooth** on that interval, which means that the function and its derivative are continuous, and only a finite number of jump discontinuities are allowed. Remember, the function has a **jump discontinuity** wherever both one-sided limits exist, but those one-sided limits aren't equivalent.

In other words, we can't have the function $f(x)$ blow up to infinity at any point in $-L \leq x \leq L$, nor can we have some infinite number of discontinuities over $-L \leq x \leq L$. Below is an example of a piecewise smooth curve. There are a finite number of jump discontinuities, and the function is continuous in between those discontinuities.

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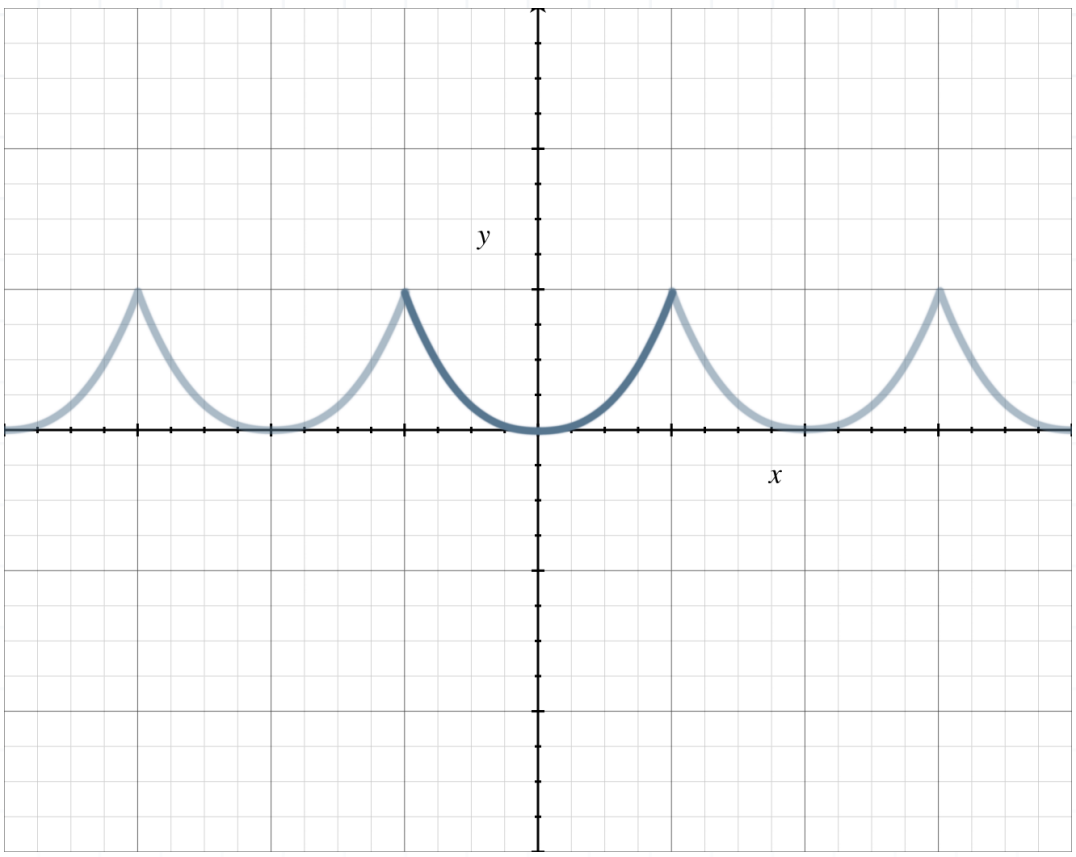



If the function is piecewise smooth, then we can take its graph on $-L \leq x \leq L$ and just repeat it over and over again to the left and right, so on $x \leq -L$ and on $x \geq L$, and the new graph that we get is the periodic extension of the original function, which we discussed in an earlier lesson. As we know, the periodic extension is a periodic function, so we can use a Fourier series to represent the periodic extension.

Then the question becomes, will the Fourier series representation of $f(x)$ converge to the periodic extension of $f(x)$? Well, if the periodic extension of $f(x)$ is continuous, then yes, the Fourier series representation will converge to $f(x)$.

For instance, since this periodic extension $g(x)$ is continuous, its Fourier series representation will converge to $g(x)$.

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On the other hand, if the periodic extension isn't continuous, and instead has a jump discontinuity between each period, then the Fourier series representation of the periodic extension $g(x)$ will converge to the average of the two one-sided limits. So if the jump discontinuity exists at $x = a$, then the Fourier series representation will converge to

$$\frac{\lim_{x\to a^-} g(x) + \lim_{x\to a^+} g(x)}{2}$$

So to summarize, if we want the Fourier series representation to converge to $f(x)$, then we need to have no jump discontinuities within the interval $-L \leq x \leq L$, and we need to make sure that $f(-L) = f(L)$.

Let's do an example where we determine the convergence of a Fourier series representation.

**Example**

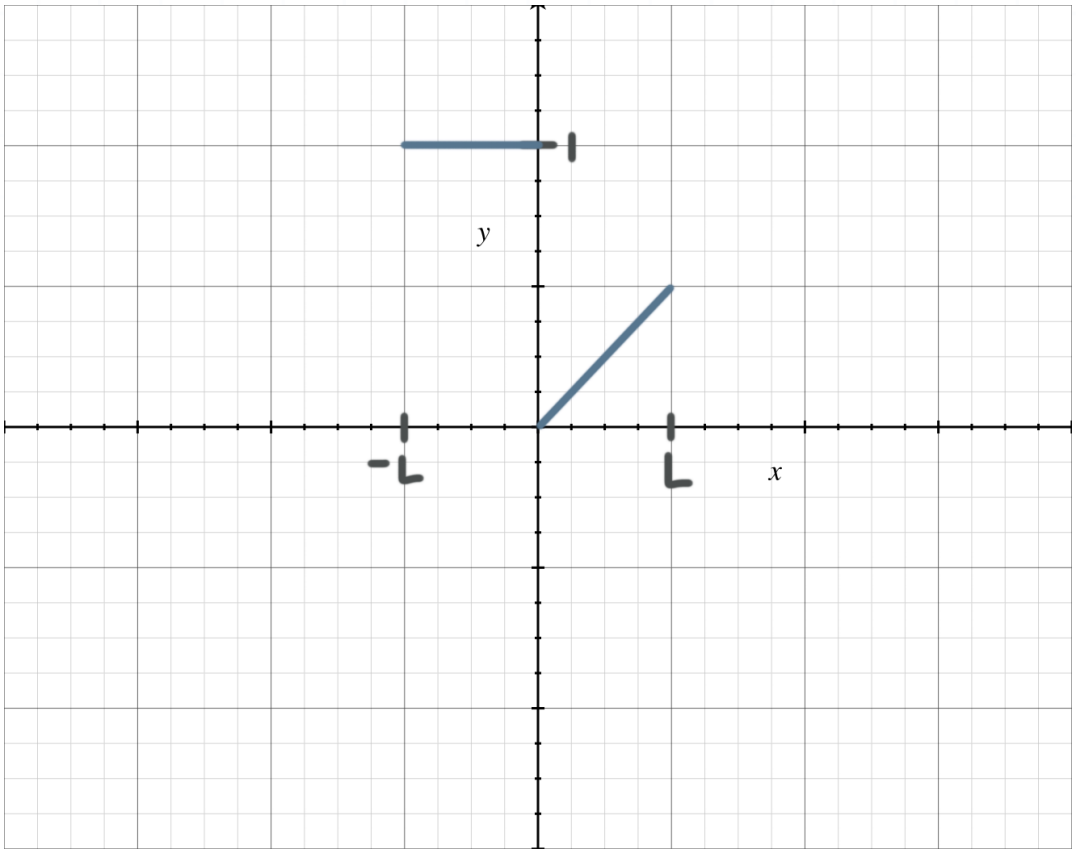
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Given the function $f(x)$ and its Fourier series representation, say whether or not the Fourier series converges to $f(x)$.

$$f(x) = \begin{cases} 1 & -L \le x < 0 \\ x & 0 \le x \le L \end{cases}$$

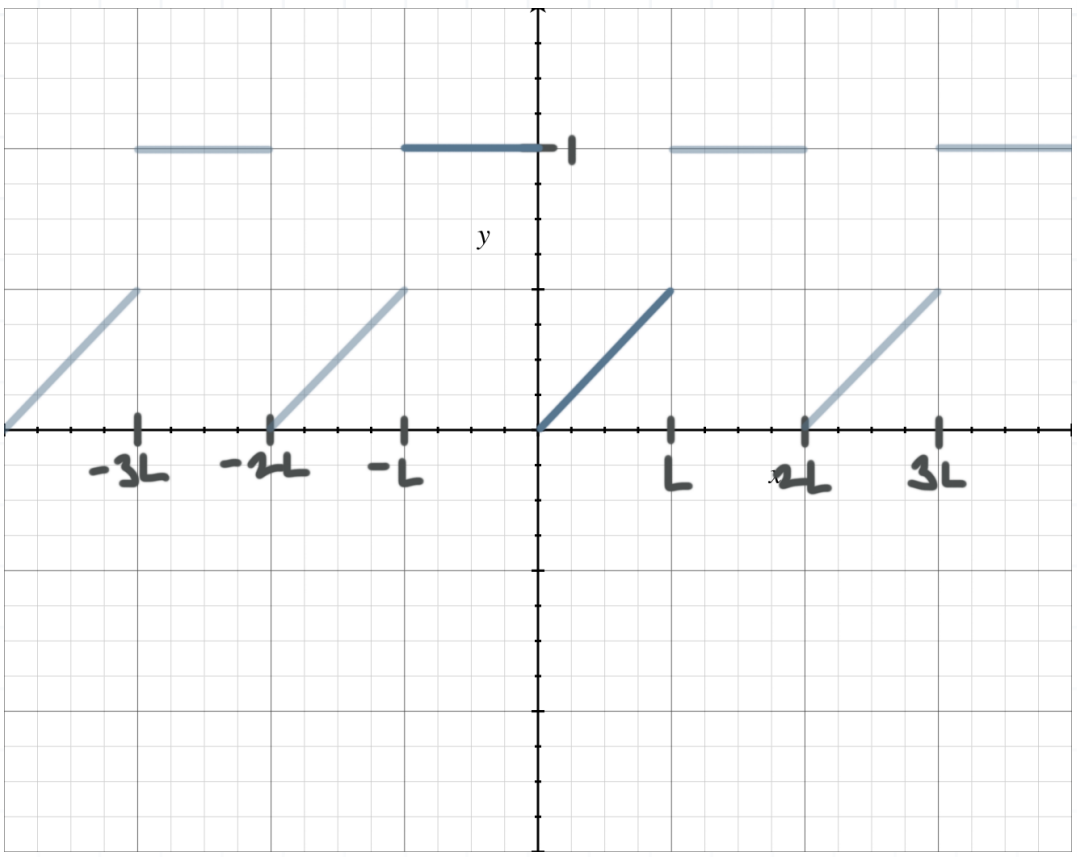
$$f(x) = \frac{L+2}{4} + \sum_{n=1}^{\infty} \frac{(-1)^n L - L}{(n\pi)^2} \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} \frac{(-1)^n (1-L) - 1}{n\pi} \sin\left(\frac{n\pi x}{L}\right)$$

This is the function we worked with in the previous lesson, where we learned to find the Fourier series representation of a piecewise defined function. A sketch of $f(x)$ is




and therefore a sketch of its periodic extension is

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From the graph of the periodic extension $g(x)$, we can see that $g(x)$ has a jump discontinuity at $x = 0$, within the interval $-L \leq x \leq L$, as well as a jump discontinuity at $-L$ and $L$, assuming $L \neq 1$.

Because the function is piecewise smooth on the individual intervals $-L \leq x < 0$ and $0 \leq x \leq L$, the function $f(x)$ and the periodic extension $g(x)$ are both continuous, so the Fourier series representation will converge to the periodic extension $g(x)$, and therefore also to the original function $f(x)$. Because of the jump discontinuity at $x = 0$, the Fourier series representation at that point will converge to

$$\frac{\lim_{x \to a^-} g(x) + \lim_{x \to a^+} g(x)}{2}$$

$$\frac{\lim_{x \to 0^-} g(x) + \lim_{x \to 0^+} g(x)}{2}$$

$$\frac{1 + 0}{2}$$

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$$\frac{1}{2}$$

Again assuming $L \neq 1$, the periodic extension will also have jump discontinuities at $x = -L$ and $x = L$. At $x = -L$, the Fourier series representation will converge to

$$\frac{\lim_{x \to a^-} g(x) + \lim_{x \to a^+} g(x)}{2}$$

$$\frac{\lim_{x \to -L^-} g(x) + \lim_{x \to -L^+} g(x)}{2}$$

$$\frac{L+1}{2}$$

$$\frac{1+L}{2}$$

and at $x = L$ the Fourier series representation will converge to

$$\frac{\lim_{x \to a^-} g(x) + \lim_{x \to a^+} g(x)}{2}$$

$$\frac{\lim_{x \to L^-} g(x) + \lim_{x \to L^+} g(x)}{2}$$

$$\frac{L+1}{2}$$

$$\frac{1+L}{2}$$

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